

Hovid-Pregabalin Capsule 150 mg

Description

White opaque and white opaque capsule with 'PB150' printed on one end and 'hovid' printed on the other end.

Composition

Pregabalin 150mg per capsule

Pharmacodynamics

Pregabalin binds to an auxiliary subunit ($\alpha_2\text{-}\delta$ protein) of voltage-gated calcium channels in the central nervous system. Efficacy has been shown in diabetic neuropathy, post herpetic neuralgia and spinal cord injury. Pregabalin has also been studied for epilepsy. A reduction in seizure frequency was observed. Pregabalin has been studied for generalized anxiety disorder (GAD). Relief of the symptoms of GAD as reflected by the Hamilton Anxiety Rating Scale (HAM-A) was observed.

Pharmacokinetics

Absorption - Pregabalin is rapidly absorbed when administered in the fasted state, with peak plasma concentrations occurring within 1 hour following both single and multiple dose administration. Pregabalin oral bioavailability is estimated to be $\geq 90\%$ and is independent of dose. Following repeated administration, steady state is achieved within 24 to 48 hours. The rate of pregabalin absorption is decreased when given with food resulting in a decrease in C_{max} by approximately 25-30% and a delay in t_{max} to approximately 2.5 hours. However, administration of pregabalin with food has no clinically significant effect on the extent of pregabalin absorption.

Distribution - The apparent volume of distribution of pregabalin following oral administration is approximately 0.56 l/kg. Pregabalin is not bound to plasma proteins.

Biotransformation - Pregabalin undergoes negligible metabolism in humans. Following a dose of radiolabelled pregabalin, approximately 98% of the radioactivity recovered in the urine was unchanged pregabalin. The N-methylated derivative of pregabalin, the major metabolite of pregabalin found in urine, accounted for 0.9% of the dose.

Elimination - Pregabalin is eliminated from the systemic circulation primarily by renal excretion as unchanged drug. Pregabalin mean elimination half-life is 6.3 hours. Pregabalin plasma clearance and renal clearance are directly proportional to creatinine clearance. Dose adjustment in patients with reduced renal function or undergoing haemodialysis is necessary.

Renal Impairment - Pregabalin clearance is directly proportional to creatinine clearance. In addition, pregabalin is effectively removed from plasma by haemodialysis (following a 4 hour haemodialysis treatment plasma pregabalin concentrations are reduced by approximately 50%). Because renal elimination is the major elimination pathway, dose reduction in patients with renal impairment and dose supplementation following haemodialysis is necessary.

Elderly - Pregabalin clearance tends to decrease with increasing age. This decrease in pregabalin oral clearance is consistent with decreases in creatinine clearance associated with increasing age. Reduction of pregabalin dose may be required in patients who have age related compromised renal function.

Indications

Neuropathic pain

Hovid-Pregabalin Capsule is indicated for the treatment of peripheral and central neuropathic pains in adults.

Epilepsy

Hovid-Pregabalin Capsule is indicated as adjunctive therapy in adults with partial seizures with or without secondary generalisation.

Generalised anxiety disorder

Hovid-Pregabalin Capsule is indicated for the treatment of Generalised Anxiety Disorder (GAD) in adults.

Contraindications

Not recommended for patients hypersensitive to the active substance or to any of the excipients.

Warning and Precautions

Diabetic patients - In accordance with current clinical practice, some diabetic patients who gain weight on pregabalin treatment may need to adjust hypoglycaemic medicinal products.

Dizziness, somnolence, loss of consciousness, confusion, and mental impairment - Pregabalin treatment has been associated with dizziness and somnolence, which could increase the occurrence of accidental injury (fall) in the elderly population.

Vision-related effects - Loss of vision, visual blurring or other changes of visual acuity. Discontinuation of pregabalin may result in resolution or improvement of these visual symptoms.

Withdrawal symptoms - insomnia, headache, nausea, anxiety, diarrhoea, flu syndrome, nervousness, depression, pain, convulsion, hyperhidrosis and dizziness, suggestive of physical dependence. The patient should be informed about this at the start of the treatment

Congestive heart failure - Pregabalin should be used with caution in these patients. Discontinuation of pregabalin may resolve the reaction.

Treatment of central neuropathic pain due to spinal cord injury - incidence of adverse reactions was increased. This may be attributed to an additive effect due to concomitant medicinal products (e.g. antispasticity agents) needed for this condition. This should be considered when prescribing pregabalin in this condition

Suicidal ideation and behaviour - Patients (and caregivers of patients) should be advised to seek medical advice should signs of suicidal ideation or behaviour emerge.

Reduced lower gastrointestinal tract function - There are reports of events related to reduced lower gastrointestinal tract function (e.g. constipation) when pregabalin was co-administered with medications that have the potential to produce constipation, such as opioid analgesics.

Misuse, abuse potential or dependence - Caution should be exercised in patients with a history of substance abuse.

Encephalopathy - patients with underlying conditions that may precipitate encephalopathy.

Pregnancy and Lactation

There are no adequate and well-controlled studies in pregnant women. Use Hovid-Pregabalin Capsule during pregnancy only if the potential benefit justifies the potential risk to the fetus. It is not known if pregabalin is excreted in human milk; it is, however, present in the milk of rats. Because many drugs are excreted in human milk, and because of the potential for tumorigenicity shown for pregabalin in animal studies, decide whether to discontinue nursing or to discontinue the drug, taking into account the importance of the drug to the mother.

Interaction with Other Medicaments and Other Forms of Interaction

Since pregabalin is predominantly excreted unchanged in the urine, undergoes negligible metabolism in humans ($< 2\%$ of a dose recovered in urine as metabolites), does not inhibit drug metabolism *in vitro*, and is not bound to plasma proteins, it is unlikely to produce, or be subject to, pharmacokinetic interactions.

Oral contraceptives, norethisterone and/or ethinyl oestradiol - Co-administration of pregabalin with the oral contraceptives norethisterone and/or ethinyl oestradiol does not influence the steady-state pharmacokinetics of either substance.

Central nervous system influencing medical products - Pregabalin may potentiate the effects of ethanol and lorazepam. Respiratory failure and coma in patients taking pregabalin and other central nervous system (CNS) depressant medicinal products have been reported. Pregabalin appears to be additive in the impairment of cognitive and gross motor function caused by oxycodone.

No clinically relevant pharmacokinetic interactions were observed between pregabalin and phenytoin, carbamazepine, valproic acid, lamotrigine, gabapentin, lorazepam, oxycodone or ethanol. Population pharmacokinetic analysis indicated that oral antidiabetics, diuretics, insulin, phenobarbital, tiagabine and topiramate had no clinically significant effect on pregabalin clearance.

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Central nervous system influencing medical products - Pregabalin may potentiate the effects of ethanol and lorazepam.

Main Side/ Adverse Effects

Infectious and infestation: Nasopharyngitis

Blood and lymphatic system disorders: Neutropaenia

Immune system disorders: Hypersensitivity, angioedema, allergic reaction

Metabolism and nutrition disorders: Appetite increased, hypoglycaemia

Psychiatric disorders: Euphoric mood, confusion, irritability, disorientation, insomnia, libido decreased, hallucination, panic attack, restlessness, agitation, depression, depressed mood, elevated mood, aggression, mood swings, depersonalisation, word finding difficulty, abnormal dreams, libido increased, anorgasmia, apathy, disinhibition

Nervous system disorder: Dizziness, somnolence, headache, Ataxia, coordination abnormal, tremor, dysarthria, amnesia, memory impairment, disturbance in attention, paraesthesia, hypoaesthesia, sedation, balance disorder, lethargy, syncope, stupor, myoclonus, loss of consciousness, psychomotor hyperactivity, dyskinesia, dizziness postural, intention tremor, nystagmus, cognitive disorder, mental impairment, speech disorder, hyporeflexia, hyperaesthesia, burning sensation, ageusia, malaise, convulsions, parosmia, hypokinesia, dysgraphia

Eye disorders: Vision blurred, diplopia, Peripheral vision loss, visual disturbance, eye swelling, visual field defect, visual acuity reduced, eye pain, asthenopia, photopsia, dry eye, lacrimation increased, eye irritation, vision loss, keratitis, oscillopsia, altered visual depth perception, mydriasis, strabismus, visual brightness

Ear and labyrinth disorders: Vertigo, Hyperacusis

Cardiac disorders: Tachycardia, atrioventricular block first degree, sinus bradycardia, congestive, heart failure, QT prolongation, sinus tachycardia, sinus arrhythmia

Respiratory, thoracic and mediastinal disorders: Dyspnoea, epistaxis, cough, nasal congestion, rhinitis, snoring, nasal dryness, pulmonary oedema, throat tightness

Gastrointestinal disorders: Vomiting, nausea, constipation, diarrhoea, flatulence, abdominal distension, dry mouth, gastroesophageal reflux disease, salivary hypersecretion, hypoaesthesia oral, Ascites, pancreatitis, swollen tongue, dysphagia

Skin and subcutaneous tissue disorders: Rash papular, urticaria, hyperhidrosis, pruritus, Stevens Johnson syndrome, cold sweat

Musculoskeletal and connective tissue disorders: Muscle cramp, arthralgia, back pain, pain in limb, cervical spasm, joint swelling, myalgia, muscle twitching, neck pain, muscle stiffness,

Renal and urinary disorders: Urinary incontinence, dysuria, renal failure, oliguria, urinary retention

Reproductive system and breast disorders: Erectile dysfunction, Sexual dysfunction, ejaculation delayed, dysmenorrhoea, breast pain, Amenorrhoea, breast discharge, breast enlargement, gynaecomastia

General disorders and administration site conditions: Oedema peripheral, oedema, gait abnormal, fall, feeling drunk, feeling abnormal, fatigue, Generalised oedema, face oedema, chest tightness, pain, pyrexia, thirst, chills, asthenia

Investigations: Weight increased, blood creatine phosphokinase increased, alanine aminotransferase increased, aspartate aminotransferase increased, blood glucose increased, platelet count decreased, blood creatinine increased, blood potassium decreased, weight decreased, White blood cell count decreased

Effects on ability to drive and use machines

Undesirable effects such as dizziness, drowsiness, fatigue and visual disturbances are possible after taking NSAIDs. If affected, patients should not drive or operate machinery

Overdose

Symptoms: Somnolence, confusional state, agitation, and restlessness

Treatment: Treatment of pregabalin overdose should include general supportive measures and may include haemodialysis if necessary.

Dosage and Administration

The dose range is 150 to 600mg per day give in either 2 or 3 divided doses. hovid-Pregabalin Capsule may be taken with or without food. If pregabalin has to be discontinued it is recommended this should be done gradually over a minimum of 1 week independent of the indication.

Neuropathic pain - Pregabalin treatment can be started at a dose of 150 mg per day given as 2 or 3 divided doses. Based on individual patient response and tolerability, the dose may be increased to 300 mg per day after an interval of 3 to 7 days, and if needed, to a maximum dose of 600 mg per day after an additional 7-day interval.

Epilepsy - Pregabalin treatment can be started with a dose of 150 mg per day given as 2 to 3 divided doses. Based on individual patient response and tolerability, the dose may be increased to 300 mg per day after 1 week. The maximum dose of 600 mg per day may be achieved after an additional week.

Generalised anxiety disorder - The dose range is 150 to 600 mg per day given as 2 to 3 divided doses. The need for treatment should be reassessed regularly.

Pregabalin treatment can be started with a dose of 150 mg per day. Based on individual patient response and tolerability, the dose may be increased to 300 mg per day after 1 week. Following an additional week the dose may be increased to 450 mg per day. The maximum dose of 600 mg per day may be achieved after additional week.

Discontinuation of pregabalin - In accordance with current clinical practice, if pregabalin has to be discontinued, it is recommended this should be done gradually over a minimum of 1 week independent of the indication

Patients with renal impairment - As pregabalin clearance is directly proportional to creatinine clearance, dose reduction in patients with compromised renal function must be individualised according to creatinine clearance (CL_{cr}), as indicated in Table 1 determined using the following formula:

$$CL_{cr} (ml/min) = \left[\frac{1.23 \times [140 - \text{age (years)}] \times \text{weight (kg)}}{\text{serum creatinine } (\mu\text{mol/l})} \right] \quad (\times 0.85 \text{ for female patients})$$

Table 1. Pregabalin dose adjustment based on renal function

Creatinine clearance (CL _{cr}) (ml/min)	Total pregabalin daily dose *		Dose regimen
	Starting dose (mg/day)	Maximum dose (mg/day)	
≥ 60	150	600	BID or TID
≥ 30 - < 60	75	300	BID or TID
≥ 15 - < 30	25 – 50	150	Once Daily or BID
< 15	25	75	Once Daily
Supplementary dosage following haemodialysis (mg)			
	25	100	Single dose*

TID = Three divided doses

BID = Two divided doses

* Total daily dose (mg/day) should be divided as indicated by dose regimen to provide mg/dose

* Supplementary dose is a single additional dose

Use in children and adolescents (12 to 17 years of age) - The safety and efficacy of hovid-Pregabalin Capsule in children below the age of 12 years and in adolescents (12-17 years of age) have not been established. Currently available data are described in SIDE EFFECTS section, PHARMACODYNAMICS and PHARMACOKINETICS but no recommendation on a posology can be made.

Elderly (over 65 years of age) population - Elderly patients may require a dose reduction of pregabalin due to a decreased renal function (see PHARMACOKINETICS).

Note: The information given here is limited. For further information, consult your doctor or pharmacist.

Storage : (Blister pack) Store below 30°C. Protect from moisture.

Presentation/ packing : (Capsule) Blister pack of 10's, 60's, 1000's

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